

In factor analysis, there are two different approaches. One is what I would call an economic factors analysis in which you actually know what the factors are, such as equity index prices, oil prices, etc. The other approach is abstract factor analysis where all you have are the returns for each security, and you process it with mathematical models and end up with a collection of abstract factors that don't necessarily coincide with economic factors. The abstract factors best describe the actual historical data you are using.

What is this type of approach called where the factors aren't specific items but rather mathematical constructs?

I call it abstract factor analysis, and principal component analysis is another version of this approach. If you have a huge collection of data points, you can find these things called principal components, which are vectors in the data space that are perpendicular to each other, and they best describe the data.

Is the perpendicularity the reason why multicollinearity is not a problem¹²?

Yes. In abstract factor analysis, you don't have that issue, but with economic factors, you can have factors that are closely related to each other.

That sounds like it is a big advantage for using abstract factors over economic factors.

Yes, I think it's the way to go. The abstract factors can often get identified with specific economic factors or linear combinations of those factors. For example, it always turns out that the biggest abstract factor, or equivalently the biggest principal component, is the stock market.

But you actually don't need that information; it is only an interesting aside. You could be completely ignorant of the real world and still apply abstract factor analysis.

That's right. We applied the factor neutrality approach, and the returns went up, and risk went way down.

What year did you make the change?